want to develop a new drug but I don't have enough money, I can get a loan from the bank, or turn to private investors and venture capital funds. When Ebola erupted in West Africa in the summer of 2014, what do you think happened to the shares of pharmaceutical companies that were busy developing anti-Ebola drugs and vaccines? They skyrocketed. Tekmira shares rose by 50 per cent and BioCryst shares by 90 per cent. In the Middle Ages, the outbreak of a plague caused people to raise their eyes towards heaven, and pray to God to forgive them for their sins. Today, when people hear of some new deadly epidemic, they pick up the phone and call their broker. For the stock exchange, even an epidemic is a business opportunity.

If enough new ventures succeed, people's trust in the future increases, credit expands, interest rates fall, entrepreneurs can raise money more easily and the economy grows. People consequently have even greater trust in the future, the economy keeps growing and science progresses with it.

It sounds simple on paper. Why, then, did humankind have to wait until the modern era for economic growth to gather momentum? For thousands of years people had little faith in future growth not because they were stupid, but because it contradicts our gut feelings, our evolutionary heritage and the way the world works. Most natural systems exist in equilibrium, and most survival struggles are a zero-sum game in which one can prosper only at the expense of another.

For example, each year roughly the same amount of grass grows in a given valley. The grass supports a population of about 10,000 rabbits, which contains enough slow, dim-witted or unlucky rabbits to provide prey for a hundred foxes. If one fox is very diligent, and captures more rabbits than usual, then another fox will probably starve to death. If all foxes somehow manage to capture more rabbits simultaneously, the rabbit population will crash, and next year many foxes will starve. Even though there are occasional fluctuations in the rabbit market, in the long run the foxes cannot expect to hunt, say, 3 per cent more rabbits per year than the preceding year.

Of course, some ecological realities are more complex, and not all survival struggles are zero-sum games. Many animals